Sylf AI Planner

Planning Intelligence Addendum

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Incremental capabilities that improve plan quality, execution clarity, and personalization without repeating the feature set already specified in the AI Planner Modes PRD v1.0.

Companion product specification. This document is intentionally additive: it defines only new planner behavior and should be read alongside the base PRD.

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1 Executive Summary

This companion addendum specifies only incremental AI Planner behavior that is not defined in the base PRD. Its focus is a narrower question: what additional intelligence would make generated plans feel substantially more realistic to execute, more academically deliberate, and more personally fitted to the student?

The additions below target three failure modes that can remain even when a plan is structurally valid: the planner may ask the wrong setup questions, a technically valid schedule may still be pedagogically weak or robotic, and a good initial plan may fail to learn from the user's corrections. The proposed features therefore improve input quality, semantic plan quality, execution quality, and preference learning.

> Additive scope rule
> No feature in this document is intended to replace or restate a capability from the base PRD. Where an addition touches an existing flow, this document specifies only the new behavior layered on top of that flow.

1.1 Recommended additions at a glance

Addition	New behavior	Primary benefit
Smart Plan Setup	Ask only the extra questions that can materially change the plan.	Higher-quality inputs with less setup friction.
Hard vs. soft constraints	Separate non-negotiable limits from preferences.	Plans fit real life without becoming brittle.
Topic priority signals	Let importance influence time allocation independently of confidence.	More time goes to what matters most.
Semantic plan critic	Run a second-pass quality review after structural validation.	Catches weak sequencing and robotic plans.
Spaced retrieval scheduler	Revisit learned material at deliberate intervals.	Better retention across the plan horizon.
Cognitive-load shaping	Balance demanding and lighter work, not only minutes.	Daily schedules feel more realistic.
Core + Stretch layers	Separate essential work from optional extra capacity.	Less overload and better recovery from variable days.
Definition of Done	Give every quest a bounded, observable completion condition.	Users know exactly when a task is finished.
Coverage Map	Show how each scope unit is touched across the plan.	Greater trust and easier gap detection.
Preference learning	Learn from repeated user corrections to planner output.	Plans become personally fitted over time.
Friction feedback	Capture optional one-tap reasons for plan-item changes.	Replanning fixes the cause, not only the date.

2 Smart Plan Setup: Value-of-Information Questioning

The setup flow should become conditionally intelligent rather than merely mode-specific. After the user supplies the minimum required information, Sylf should identify the small number of unknowns most likely to change sequencing, workload, or prioritization. It should ask only those questions and skip low-value questions entirely.

2.1 Question-selection policy

- Generate at most 1-3 additional questions by default. A question is justified only when different answers would materially change the resulting plan.
- Prefer closed or low-effort choices when possible. Free text should be used only when the missing information cannot be captured cleanly.
- Each question may expose a short "Why this matters" hint so the user understands why Sylf is asking.
- If the planner already has enough information to produce a high-quality plan, it should proceed without manufacturing extra setup steps.
- Questions must be derived from planning uncertainty, not from curiosity or a desire to collect more user data.

2.2 Examples of high-value questions

Detected uncertainty	Example question
Two topics have equal low confidence but very different importance.	"Which of these matters more for this assessment?"
The user gives weekly hours but no usable distribution pattern.	"Would you rather spread study evenly or keep longer sessions on fewer days?"
A preferred study window conflicts with the only feasible schedule.	"Is this study window a preference or a hard limit?"
A topic label is too broad to size reliably.	"Is this one chapter, several subtopics, or the full unit?"

2.3 Acceptance criteria

- The planner can skip all optional setup questions when no answer would materially change the plan.
- Every generated question has a machine-readable reason code tied to a planning decision.
- Question count is bounded and cannot expand into an interview-like flow without explicit user choice.
- Changing an answer produces a measurable change in the affected plan dimension or the question should not have been asked.

3 Hard vs. Soft Constraints

User constraints should be typed by strength. A hard constraint is a rule the scheduler must never violate. A soft constraint is a preference the planner should optimize for but may break when necessary to preserve a more important goal. This distinction prevents two common failures: silently breaking a real-life limit, or declaring a plan impossible because a preference was treated as absolute.

Type	Example	Planner behavior
Hard	"I cannot study on Friday."	Never schedule work there.
Hard	"No session longer than 60 minutes."	Split or move work; do not exceed the limit.
Soft	"I prefer maths in the evening."	Prefer evening placement, but allow another time if needed.
Soft	"Avoid two heavy sessions back-to-back."	Optimize around the preference; explain a necessary exception.

3.1 New data requirements

- Constraint strength: HARD or SOFT.
- Optional soft-preference weight or rank when several preferences conflict.

- Source: explicitly supplied, inferred from a user edit, or learned preference.
- Violation explanation for any soft constraint the final plan cannot satisfy.

3.2 Acceptance criteria

- Hard constraints are validated deterministically before a plan can be shown.
- Soft constraints influence scheduling but never cause a false infeasibility result on their own.
- When a soft preference is violated, the draft can explain the tradeoff in one concise sentence.

4 Topic Priority Signals

Confidence answers "how well do I know this?" but not "how much does this matter?" The planner should support an independent topic-priority signal so that a high-importance topic can receive more attention even when the student is moderately confident, while a low-importance weak topic does not automatically consume disproportionate time.

Priority may come from explicit user input such as assessment weighting, teacher emphasis, a must-cover flag, or a personal priority. Sylf should not invent official weighting. If a priority is not user-supplied or verified, it should remain neutral.

4.1 Allocation behavior

- Use priority and confidence as separate inputs. Neither should overwrite the other.
- High-priority weak topics receive earlier and deeper treatment when capacity allows.
- High-priority strong topics may receive shorter reinforcement rather than full relearning.
- Low-priority weak topics may receive bounded coverage when the deadline is tight, with the tradeoff visible to the user.
- Priority must influence practice allocation as well as initial learning time.

> Example
> A student rates Topic A as 2/5 confidence but low assessment importance, and Topic B as 3/5 confidence but very high importance. A confidence-only planner may over-invest in Topic A. Priority-aware planning can protect meaningful practice for Topic B without pretending the student already masters it.

5 Semantic Plan Critic and Human Review Pass

A plan can pass every structural validator and still be poor. Dates can be legal, capacity can fit, and every topic can appear - yet the sequence may feel repetitive, the task wording may be vague, or the plan may place several demanding activities in an awkward pattern. Sylf should therefore add a semantic quality gate after structural validation and before the draft is presented.

5.1 Critic responsibilities

- Detect pedagogically weak sequencing that is not expressible as a simple deterministic rule.
- Detect redundant or near-duplicate plan items that are technically different but functionally repetitive.
- Flag vague quests whose action or expected output is unclear.
- Detect unnatural task density, abrupt difficulty jumps, or repeated high-effort patterns that make the plan feel machine-generated.
- Check whether the plan meaningfully uses the user's stated priorities and soft preferences.
- Check whether retrieval opportunities are distributed rather than clustered.

5.2 Repair policy

1. The critic returns issue codes, affected item IDs, severity, and a concise patch recommendation. It does not regenerate the whole plan by default.
2. Only the affected items are repaired.
3. The repaired segment is run through structural validation again.
4. If the critic and scheduler disagree, deterministic hard constraints remain authoritative.
5. The user never sees internal chain-of-thought; only the improved draft and, when useful, a concise user-facing explanation.

5.3 Suggested issue codes

Code	Meaning
SEQ_WEAK	A prerequisite or conceptual progression is awkward.
TASK_VAGUE	The user cannot tell what concrete work to perform.
DENSITY_SPIKE	A day is technically within time limits but cognitively overloaded.
REDUNDANT_TOUCH	Two items repeat the same learning purpose without justification.
PRIORITY_MISS	The plan does not reflect a supplied high-priority topic.
ROBOTIC_PATTERN	Wording or task pattern is unnaturally repetitive.

6 Spaced Retrieval Scheduler

Sylf should deliberately schedule later retrieval of material after initial learning instead of allowing all review to cluster near the end. This addition does not create a new quest type; it changes the timing and recurrence policy of existing recall/review work.

6.1 Scheduling behavior

- After a topic is first learned or substantially reviewed, create one or more short retrieval touches later in the horizon when time permits.
- Spacing should scale with the remaining horizon. A 7-day plan and a 60-day plan should not use the same fixed intervals.
- Retrieval touches should be brief and specific: recall key ideas, solve a small representative set, or reproduce a method without notes.
- If the user demonstrates that a topic is already secure, later retrieval can be shorter rather than removed automatically.
- When the deadline is extremely short, the scheduler may compress spacing but should avoid placing all recall immediately after learning.

6.2 Planner representation

Field	Purpose
retrievalGroupId	Links later retrieval touches to the source topic or learning item.
retrievalIndex	1, 2, 3... within the planned sequence.
minimumGap	Deterministic minimum separation when the horizon permits.
retrievalPurpose	Recall, representative practice, or final reinforcement.

7 Cognitive-Load Shaping

Minutes alone are not a complete measure of study load. Forty minutes of timed problem solving can be more demanding than forty minutes of light review. Each plan item should therefore carry a coarse cognitive-intensity label that the scheduler uses to shape a realistic day.

7.1 Intensity model

Intensity	Typical use
LOW	Light recall, short review, organizing errors, quick reinforcement.
MEDIUM	Normal learning, guided examples, standard topic practice.
HIGH	New difficult concepts, timed practice, mocks, dense problem solving.

7.2 Scheduling rules

- Avoid stacking multiple HIGH items on the same day when equivalent scheduling options exist.
- Use LOW or MEDIUM work around demanding assessment practice to create a more executable daily rhythm.
- A day may be within its minute limit and still be flagged by the critic if its intensity mix is unrealistic.
- Intensity is an academic workload label, not an inference about health, disability, mood, or medical capacity.

8 Core + Stretch Quest Layers

Daily plans should distinguish essential work from optional extra capacity. The planner may label a small number of tasks as Core and additional work as Stretch. Stretch work is useful when the student has extra time, but skipping it should not create an overdue-task cascade.

8.1 Behavior

- Core work contains the highest-value tasks needed to keep the plan on track.
- Stretch work uses remaining safe capacity for extra practice, reinforcement, or getting ahead.
- Stretch items are never presented as required to maintain the plan unless a later revision explicitly promotes them to Core.
- When a user has a short day, the UI can surface the Core item first rather than showing a wall of equally urgent quests.
- If Stretch work is completed, the next revision may reduce redundant future reinforcement rather than simply adding more work.

> Product effect
> This gives Sylf a graceful answer to variable real-life days: the plan can remain ambitious without making every optional minute feel like a failed obligation.

9 Quest Definition of Done

Every actionable planner quest should include a concise, observable completion condition. A title such as "Practice electric fields" is directionally useful but leaves the user to decide how much practice is enough. The planner should specify the intended output without turning each quest into a long instruction sheet.

9.1 Examples

Quest	Definition of Done example
Review induction	Recall the core laws from memory, then check and correct gaps.
Practice capacitors	Complete one bounded question set and review every incorrect step.
Mock exam	Complete the selected paper under the planned time limit, then log the main error categories.

9.2 Rules

- Completion conditions must be bounded and realistically achievable within the item's planned duration.
- They should describe an observable output or action, not a subjective claim such as "fully understand".
- They should not invent official question counts, mark schemes, or exam requirements unless those were supplied or verified.
- The UI should keep the completion condition secondary to the quest title so the planner remains easy to scan.

10 Syllabus Coverage Map

The planner should expose a compact coverage map that lets the user inspect how every supplied scope unit is represented across the plan. This is more than a progress percentage: it is a traceability view from topic to planned learning touches.

10.1 Topic states

State	Meaning
UNSCHEDULED	No plan item currently touches the scope unit.
SCHEDULED	At least one learning/review item exists.
REINFORCED	The topic has a later retrieval/review touch.
PRACTICED	The plan includes application/practice for the topic.

10.2 UX behavior

- Selecting a topic shows every plan item that touches it and the purpose of each touch.
- Orphaned topics are visually obvious before commit.
- The map must not label a topic "mastered" solely because it appears in the schedule.
- For large syllabi, the map should collapse by chapter/unit and expand on demand.

11 Planner Preference Learning from User Corrections

Sylf should learn from repeated explicit corrections to planner output. The goal is not to infer a broad personality profile; it is to remember narrow planning preferences that the user repeatedly demonstrates so future drafts require fewer edits.

11.1 Learnable planner preferences

- Preferred practical session length when the user repeatedly shortens generated durations.
- Preferred daily quest density when the user repeatedly merges or removes small items.

- Preferred placement of demanding work when the user repeatedly moves similar items to the same time/day pattern.
- Preferred level of task detail when the user consistently expands or simplifies planner-generated instructions.
- Preferred buffer style, such as fewer explicit catch-up tasks versus more unallocated space, when the user repeatedly edits one pattern into the other.

11.2 Learning safeguards

- Learn only from repeated or explicitly confirmed behavior; a single edit should not become a permanent rule.
- Keep planner preferences separate from hard availability and academic facts.
- Surface a learned preference when it materially changes a new plan, with a simple way to ignore or reset it.
- Do not infer sensitive traits, health conditions, motivation, intelligence, or capability from planning edits.
- Preference learning must never silently convert a soft pattern into a hard constraint.

11.3 Example

> Preference adaptation
> If a user repeatedly changes 60-minute generated quests to 40 minutes, Sylf can begin sizing future items around 40 minutes and split larger tasks earlier. This reduces repetitive correction without claiming that 40 minutes is objectively optimal.

12 Friction Feedback Loop

When a user removes, postpones, shortens, or repeatedly avoids a planner item, Sylf should optionally ask for a one-tap reason. This gives replanning a causal signal. Without it, the system may keep moving the same poorly designed task to a new date.

12.1 Suggested feedback reasons

Reason	Planner interpretation
TOO_LONG	Task was larger than the available time.
TOO_HARD	Task needed more preparation or a smaller step.
UNCLEAR	The user did not know what to do.
NO_MATERIAL	Required notes, paper, or resource was unavailable.
ALREADY_KNEW	The planned work was redundant.
LOW_VALUE	The user did not consider the item worth the time.

12.2 Response policy

- Feedback is optional and should take one tap in the common case.
- TOO_LONG should encourage smaller future task sizing rather than merely moving the same task.
- TOO_HARD should encourage prerequisite work or a smaller first step.
- UNCLEAR should trigger a clearer Definition of Done.
- ALREADY_KNEW should reduce redundant learning and preserve only useful reinforcement.

- A single feedback event affects the current revision; repeated patterns may feed the planner preference profile.

13 New Data and Orchestration Additions

The following additions are the minimum new planner concepts needed to support this addendum. They are intentionally expressed as conceptual fields rather than a mandatory database schema.

Concept	New fields / purpose
PlannerConstraint	strength, preferenceWeight, source, violationReason
PlannerScopeItem	prioritySignal, prioritySource
PlannerItem	cognitiveIntensity, coreStretchRole, completionCriteria, retrievalGroupId, retrievalIndex
PlannerPreference	preferenceType, value, evidenceCount, confidence, lastConfirmedAt
PlannerFrictionEvent	planItemId, reasonCode, optional note, revisionId
PlanCriticIssue	issueCode, affectedItemIds, severity, patchInstruction

13.1 Updated generation sequence

1. Collect the minimum mode inputs.
2. Run Smart Plan Setup and ask only high-value missing questions.
3. Resolve hard constraints, soft preferences, and topic-priority signals.
4. Generate the structured plan.
5. Apply spaced-retrieval timing, cognitive-load shaping, and Core/Stretch labeling.
6. Generate Definition of Done text for actionable items.
7. Run existing structural validation.
8. Run the semantic plan critic and patch only flagged segments.
9. Run structural validation again on patched segments.
10. Build the Coverage Map and present the draft.

13.2 Model separation recommendation

The generator and critic should use separate prompts and separate structured schemas even if the same provider/model is used. The critic should receive the normalized inputs and generated plan, but it should not be allowed to rewrite hard constraints or create persisted identifiers. This separation makes semantic QA testable and reduces the chance that a "review" step simply regenerates the same mistakes.

14 New Analytics and Quality Metrics

Metrics for these additions should measure whether the planner needs fewer corrections and produces more executable plans. Raw private study content should not be required to calculate these metrics.

Metric	Purpose
smart_question_asked / skipped	Whether conditional setup reduces unnecessary questions.
soft_constraint_satisfied_rate	How often plans honor user preferences when feasible.
critic_issue_rate	Which semantic defects are common before the draft reaches the user.

Metric	Purpose
critic_patch_success_rate	Whether repaired segments pass structural validation and avoid repeat flags.
planner_item_user_edit_rate	Whether new plan quality reduces manual corrections.
definition_of_done_edit_rate	Whether completion criteria are useful or frequently rewritten.
stretch_completion_rate	Whether optional work is useful without becoming assumed capacity.
friction_reason_distribution	Which planner-generated task problems users encounter most often.
learned_preference_override_rate	Whether learned preferences actually match the user.

15 Acceptance Criteria

- Smart Plan Setup asks no optional question unless the answer can materially affect a planning decision.
- Hard constraints cannot be violated by generation, critic repair, or later personalization.
- Soft preferences can be traded off without falsely making the plan infeasible, and material violations are explainable.
- Topic priority is represented independently from confidence and affects allocation when capacity is constrained.
- Every displayed draft has passed both structural validation and the semantic quality gate.
- The semantic critic patches affected segments rather than regenerating the entire plan by default.
- Plans with sufficient horizon deliberately separate at least some retrieval from initial learning.
- Daily scheduling accounts for cognitive intensity in addition to total minutes.
- Stretch work can be skipped without becoming an automatic overdue obligation.
- Every actionable planner quest has a bounded, observable Definition of Done.
- The Coverage Map can trace each supplied scope unit to its scheduled learning touches and can expose unscheduled units.
- Learned planner preferences require repeated evidence or explicit confirmation and remain user-controllable.
- Friction feedback is optional, fast, and can alter the cause of a future plan item rather than only its date.

16 Recommended Implementation Priority

Phase	Additions	Why this order
A1 - Plan quality foundation	Smart Plan Setup; hard/soft constraints; topic priority; Definition of Done; semantic critic.	Highest immediate effect on first-plan satisfaction and trust.
A2 - Learning and execution quality	Spaced retrieval; cognitive-load shaping; Core/Stretch; Coverage Map.	Makes plans easier to execute and academically stronger.
A3 - Personalization loop	Preference learning; friction feedback.	Reduces repeated edits and improves later revisions over time.

The strongest near-term combination is A1. It improves the quality of the very first generated plan before relying on historical user behavior. A2 then makes the schedule more realistic and retention-aware. A3 should follow once enough planner interactions exist to learn preferences safely and usefully.

17 Product Decision Principles for These Additions

1. Ask fewer questions, but make each question matter.
2. Treat user limits as constraints and user habits as preferences unless the user explicitly says otherwise.
3. A structurally valid plan is not automatically a good plan; semantic quality needs its own gate.
4. Optimize for executable days, not maximum scheduled minutes.
5. Make every quest concrete enough that the student can start without deciding what the planner meant.
6. Use repetition deliberately through spacing, not through accidental duplicate tasks.
7. Learn from user corrections without turning the planner into an opaque behavioral profile.
8. When the planner receives feedback, fix the underlying task design where possible instead of merely moving the task.
9. Expose enough traceability for the user to trust the plan without overwhelming the main planning view.

End of Addendum - Sylf AI Planner Planning Intelligence - Draft v1.0